

$$i_D = k'_n \frac{W}{L} \left[(v_{GS} - V_t) v_{DS} - \frac{1}{2} v_{DS}^2 \right]$$

$$i_D = \frac{1}{2} k'_n \frac{W}{L} (v_{GS} - V_t)^2$$

$$g_m = \frac{\partial i_D}{\partial v_{GS}} \big|_{v_{GS}=V_{GS}} = k'_n \frac{W}{L} (V_{GS} - V_t)$$